A FINITE GROUP ACTS TRANSITIVELY ON THE PRIMES OF AN INTEGRAL EXTENSION LYING OVER A FIXED PRIME OF THE INVARIANT SUBRING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Integrality of the invariant subring extension, via the monic polynomial which is the product over the group of the conjugates. used at proof lines 32687-32689. Is the integrality input on which the no-inclusion criterion is applied.
- The absence of inclusion relations among primes over a given prime in an integral extension. invoked at proof lines 32690-32691. Locates the primes over p so that conjugacy is the only remaining question.
- Prime avoidance. invoked at proof lines 32693-32694. Produces the element x used to build the invariant y.

Source reference. Original lemma 00GT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 619–619. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00DS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 508–508. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let G be a finite group acting on a ring R. For any two primes $\mathfrak{q}, \mathfrak{q}' \subset R$ lying over the same prime in R^G there exists a $\sigma \in G$ with $\sigma(\mathfrak{q}) = \mathfrak{q}'$.*

Proof. The extension $R^G \subset R$ is integral because every $x \in R$ is a root of the monic polynomial $\prod_{\sigma \in G}(T - \sigma(x))$ in $R^G[T]$. Thus there are no inclusion relations among the primes lying over a given prime $\mathfrak{p}$ (Algebra, Lemma 00GT). If the lemma is wrong, then we can choose $x \in \mathfrak{q}'$, $x \notin \sigma(\mathfrak{q})$ for all $\sigma \in G$. See Algebra, Lemma

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

00DS. Then $y = \prod_{\sigma \in G} \sigma(x)$ is in R^G and in $\mathfrak{p} = R^G \cap \mathfrak{q}'$. On the other hand, $x \notin \sigma(\mathfrak{q})$ for all σ means $\sigma(x) \notin \mathfrak{q}$ for all σ. Hence $y \notin \mathfrak{q}$ as $\mathfrak{q}$ is a prime ideal. This is impossible as $y \in \mathfrak{p} \subset \mathfrak{q}$. $\square$